

Welcome to Code Crunch!

C11 | CRUNCH ARCADE

[Tuesdays 3-4 PM]/[Wednesdays
10-11 AM] Weekly | Spring 2025

Attendance



linktr.ee/CODE.CRUNCH





Unit #1

C[11] | [CRUNCH ARCADE]

Introduction to Crunch Arcade and Unity



Welcome to Code Crunch!

Code Crunch is your go-to platform for **collaboration**, **networking**, and **hands-on projects**. We welcome freshmen, graduates, and alumni from **any university**, regardless of **major** or **experience level**.

- Join us **virtually via Zoom** or **in-person at Florida International University**. Together, we'll break down barriers in technology, learn through immersive **workshops**, and build **innovative solutions**.



Our Mission & Vision

Mission:

- Our **mission** is to **equip students** with the **technical skills** and **professional connections** needed to excel in **coding interviews** and beyond. We provide a platform where students can **collaborate**, **learn**, and **grow professionally** while emphasizing the importance of **diversity** and **inclusion** in the tech industry vision.
- We strive to bridge the gap between **academia** and the **tech industry** by offering **hands-on workshops**, **technical interview preparation**, **hackathons**, and **networking opportunities** with experts from **leading companies**. Our **vision** is to empower students from all backgrounds to achieve their **full potential**, **foster** an **inclusive tech community**, and build a strong, supportive alumni network. By covering a wide range of technical areas, we aim to prepare students for success in a variety of tech careers.



What We Offer

- **Hands-On Workshops:** Topics include Python, JavaScript, React, and more.
- **LeetCode Prep:** Weekly sessions to master technical interviews.
- **Hackathons:** Collaborate on innovative projects in a competitive setting.
- **Panel Discussions:** Industry leaders from tech giants and startups share insights.
- **Global Access:** Code Crunch Global launches in October for worldwide participation!

Meet the E-Board

- **President:** Brian Bazurto
- **Vice President:** Amilcar Pena
- **Treasurer:** Bryan Garcia
- **Marketing/Logistics:** Mariana Phelan

Tech. Engagement Chairs:

- | | |
|-----------------------|--------------------|
| ● Amilcar Pena | ● Thierry Laguerre |
| ● Daniel Hernandez | ● Johan Gonzalez |
| ● Arnold Miranda-Pino | ● Ashley Prado |
| ● Sara Salvatierra | ● Gabriela Machado |
| ● Gabriel Nogueras | ● Jason Jiang |
| ● Justin Joseph | ● Henry Cruz |





Why Join Code Crunch?

- **Collaborate** with peers and mentors.
- **Develop** technical and leadership skills.
- **Network** with professionals from top companies.
- **Build** impactful projects and gain hands-on experience.

A welcoming and inclusive space for underrepresented groups in tech.

Spring 2025 Weekly Schedule





Funded by SGA | Open to All Students

CODE CRUNCH WEEKLY SCHEDULE - SPRING 25

Your CS Club for Collaboration, Networking, and Hands-On Projects!
Open to all students, regardless of major—no experience needed! **Freshmen, graduates, and alumni from any university are welcome.** Let's innovate together!

C5 Crunch AI Data Science Python MON 1 - 2 PM WED 1 - 2 PM	C3 CodeCrunch LABS Portfolio Java & React MON 2 - 3 PM	C7 Wire Crunch: Embedded Systems C++ TUE 1 - 3 PM	C11 Crunch Arcade Unity TUE 3 - 4 PM WED 10 - 11 AM	C1 Crunch Convos Python WED 2 - 3 PM	C6 Cybersecurity Crunch VM/LINUX THU 1 - 2 PM
C4 CodeCrunch LABS Hackathon C++ THU 2 - 3 PM	C8 CodeCrunch LABS: Web Dev C++ THU 3 - 4 PM	C2 Crunch The Code C++ FRI 1 - 2 PM	C9 Swe Convos C++ FRI 2 - 3 PM	C10 Crunching Bioinformatics C++ FRI 3 - 4 PM	

Attend 7 of 10 Weekly Sessions in the Same Workshop to Earn a Certificate of Completion!

RSVP on
LU.MA/CODECRUNCH

In person or
Stream on
Zoom





CODECRUNCH.CS.FIU.EDU

January 6, 2025 - April 4, 2025
Cost: No-cost

Become a member

Be Part of the Code Crunch Community



1. **RSVP** for sessions at go.fiu.edu/codecrunch.
2. **Follow us** on LinkedIn, Instagram, and Discord for updates.
3. **Become a member** using the QR code on this slide.
4. **Attend** workshops, **participate** in events, and **join** our leadership team.
Together, we innovate and grow.

What we offer at C11 Crunch Arcade



- This **7-week Unity course** is designed to take you from a **complete beginner** to a **confident game developer** with a **fully functional game** of your own.
- We will **guide** you through the **essential skills** needed to bring your ideas to life, including **game design principles**, **C# scripting**, **UI development**, **environment creation**, **AI implementation**, and **game polishing techniques**.
- By the **end** of this course, you will not only have a **finished project** to showcase but also a **solid understanding** of Unity's tools and workflows.
- Whether you dream of **creating** your own indie game, **breaking** into the **game development industry**, or **simply exploring** your creativity, this course will give you the foundation you need to get started!

Key Dates and Structure

C11 - Crunch Arcade



★ Week 1: Introduction to Unity & Game Development

Learning what Unity is, installing Unity Engine and necessary modules for setup.

★ Week 2: Unity Components & Scripting Basics

Understanding physics, collisions, and writing your first C# scripts.

★ Week 3: UI & Playable Character

Creating UI elements, adding score systems, and implementing player movement.

★ Week 4: Environment Design & Interaction

Designing terrain, adding collectibles, and building game logic.

★ Week 5: AI & Gameplay Mechanics

Introducing enemies, adding combat, and designing levels.

★ Week 6: Sound, Visual Effects & Testing

Implementing audio, particle effects, and playtesting for final adjustments.

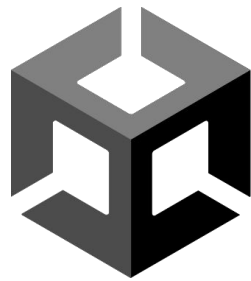
★ Week 7: Final Project Showcase

Presenting your game, sharing your experience, and receiving feedback.



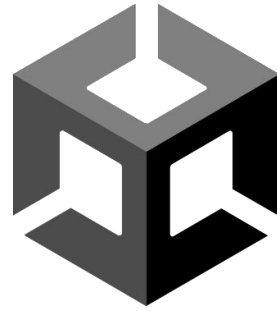
Starting off...

Let's Get Started! :D



Unity

Installations!



Unity



- Let's start off first by downloading **Unity** and its **necessary components**.
- Head to: <https://unity.com/download>, and select either download for Windows, Mac or Linux depending on your system.
- Then make a **free account**, and install the Unity Editor.
- After installing, go to “**Your Installs**” and select the  icon (on the right) and click on “**Add modules.**”
- Select “**Microsoft Visual Studio Community 2022**” -- this will be used for **scripting**.
 - Within the installation wizard, scroll down to the “**Gaming**” section and choose: “**Game development with Unity**” and hit the install button.



While that's installing...

**What is Unity exactly,
and why?**





Unity Lore and Overview

What is Unity?

- A cross-platform game engine developed by Unity Technologies.
- It is mainly used for creating 2D, 3D, AR, and VR applications.

Background Information:

- First released in 2005 as an Apple-exclusive engine.
- Expanded to support multiple platforms, now a leading game engine.
- Uses C# as its primary programming language.

What is it Used For?

- Game development (PC, mobile, console, AR/VR).
- Real-time simulations, architectural visualization, film, and interactive media.

Industry Impact:

- Powers over 50% of mobile games and nearly half of all games.
- Used by indie developers and major studios alike.
- Integral to emerging technologies like AR/VR and metaverse applications.

Key Concepts in Game Development:

- **GameObjects & Components** – The building blocks of a Unity game.
- **Scripting (C#)** – Controls gameplay mechanics and interactions.
- **Physics & Collisions** – Enables movement, gravity, and interactions.
- **UI & User Interaction** – Menus, buttons, and player feedback.



End of Unit 1!

That's **end** of today's
workshop.



Q&A Session



Attendance



Join us for the upcoming weekly sessions this Spring 2025!
(Jan 6 - Apr 4, 2025)



Monday: C5 1PM / C3 2PM
Tuesdays: C7 1PM
Wednesdays: C5 1PM / C1 2PM
Thursdays: C6 1PM / C4 2PM / C8 3PM
Friday: C2 1PM / C9 2PM / C10 3PM



RSVP [LU.MA/CODECRUNCH](https://lu.ma/codecrunch)



Your feedback helps us improve. Share your thoughts!
Go to our website: go.fiu.edu/codecrunch